

Phase 8: MongoDB Pool + ALB Readiness

Previous backend behavior compared with the completed Phase 8 build

CODE STATUS	VERIFICATION	DEPLOYMENT
Complete	27 passed / 0 failed	Ready - not deployed

Executive summary

Phase 8 adds explicit MongoDB connection-pool and timeout controls, plus a lightweight **GET /api/ready** endpoint for a future AWS ALB target group. The route verifies real MongoDB and Redis/Valkey connectivity with read-only pings, returns HTTP 503 when either critical dependency is unavailable, and discloses no raw provider errors or connection details.

Previous version vs Phase 8

Area	Previous version	Phase 8 build
MongoDB startup	<code>mongoose.connect(MONGODB_URI)</code> used driver defaults.	Startup supplies four explicit, environment-controlled options with safe positive-integer parsing and production defaults.
Connection logging	A failed startup logged the raw error message.	Failure logging is reduced to a generic message plus a safe uppercase error code, preventing URI or credential leakage.
Liveness	GET /api/health returned process uptime and cache status.	Behavior and response shape are preserved unchanged.
Readiness	No ALB-oriented endpoint. The older DB diagnostic route performs course counts and does not check Redis.	GET /api/ready checks Mongo readyState, performs an admin ping, and PINGs the existing Redis singleton. No counts or writes.
Failure policy	No combined MongoDB + Redis readiness decision.	HTTP 200 only when both dependencies respond; otherwise HTTP 503 with connected/disconnected labels only.

Effective MongoDB defaults

Environment variable	Mongoose option	Default
MONGODB_MAX_POOL_SIZE	maxPoolSize	20
MONGODB_MIN_POOL_SIZE	minPoolSize	2
MONGODB_SERVER_SELECTION_TIMEOUT_MS	serverSelectionTimeoutMS	5000 ms
MONGODB_CONNECT_TIMEOUT_MS	connectTimeoutMS	10000 ms

Readiness contract

MongoDB	Redis / Valkey	HTTP	ok
connected	connected	200	true
disconnected	connected	503	false
connected	disconnected	503	false
disconnected	disconnected	503	false

Verification performed

Check	Result	Evidence
Focused Phase 8 suite	PASS	11/11 Node test records, comprising the parent group and 10 requested behavioral subtests.
Adjacent cache + locks + readiness	PASS	27/27 tests; 0 failed, cancelled, skipped, or todo.
JavaScript syntax	PASS	server.js, MongoDB config, and readiness service passed node --check.
No writes	PASS	MongoDB admin ping and Redis PING only; no database/cache mutation paths in readiness.
Singleton reuse	PASS	Two readiness calls created one Redis client, connected once, and issued two PINGS.
Secret redaction	PASS	Injected credential-bearing MongoDB and Redis errors did not appear in response bodies or logs.

Files changed for Phase 8

- **edunex-b/server.js** - wires pool options and the new readiness handler; sanitizes MongoDB startup failure logging.
- **edunex-b/.env.example** - documents the four MongoDB pool and timeout variables.
- **edunex-b/config/mongodb.js** - adds defaults and safe positive-integer parsing.
- **edunex-b/services/readinessService.js** - implements read-only dependency checks and the stable response contract.
- **edunex-b/tests/readiness.test.cjs** - covers all ten requested Phase 8 test behaviors.

Scope controls and deployment note

No PhonePe, Razorpay, Google Play, Apple billing, rate-limiting logic, Redis cache semantics, cron-lock logic, thumbnail storage, auth, OTP, frontend, PM2, DNS, ALB, or AWS infrastructure was modified by Phase 8. Separate pre-existing Phase 7 working-tree changes were preserved and are not part of this comparison. No deployment was attempted.

READY TO DEPLOY: YES

Deployment remains a separate, manual action.